Analysis of K2-18b: A Potentially Habitable Exoplanet

Project Report Submitted in fulfillment of the Requirements for the Award of the Internship of Winter Training Program on Space Science Technology

Subject Name: Astronomy and Astrophysics

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Abstract:

This project presents a comprehensive analysis of K2-18b, a sub-Neptune exoplanet orbiting an M-dwarf star located 124 light-years from Earth. Using time-series photometric data from the NASA Exoplanet Archive comprising 3,551 observational data points, I successfully extracted and processed light curve data through systematic detrending and normalization techniques. The analysis included calculating fundamental planetary parameters such as radius, equilibrium temperature, and density using established astrophysical equations, followed by manual identification and measurement of transit events to determine the transit depth and validate the planet's orbital characteristics.

My findings validate with the published parameters for K2-18b, confirming a mass of 8.63 Earth masses, radius of 2.61 Earth radii, and a density of 2.67 g/cm³—intermediate between rocky Earth-like planets and gaseous ice giants. The measured transit depth of approximately 0.276% aligns with literature values, while the calculated equilibrium temperature places K2-18b firmly within the habitable zone of its host star. Recent spectroscopic observations by the James Webb Space Telescope have detected water vapor, methane, and carbon dioxide in the planet's atmosphere, with tentative evidence for dimethyl sulfide—a potential biosignature—though this remains controversial and requires additional high-confidence observations. This analysis demonstrates that K2-18b represents a compelling target for continued atmospheric characterization and astrobiological investigation.

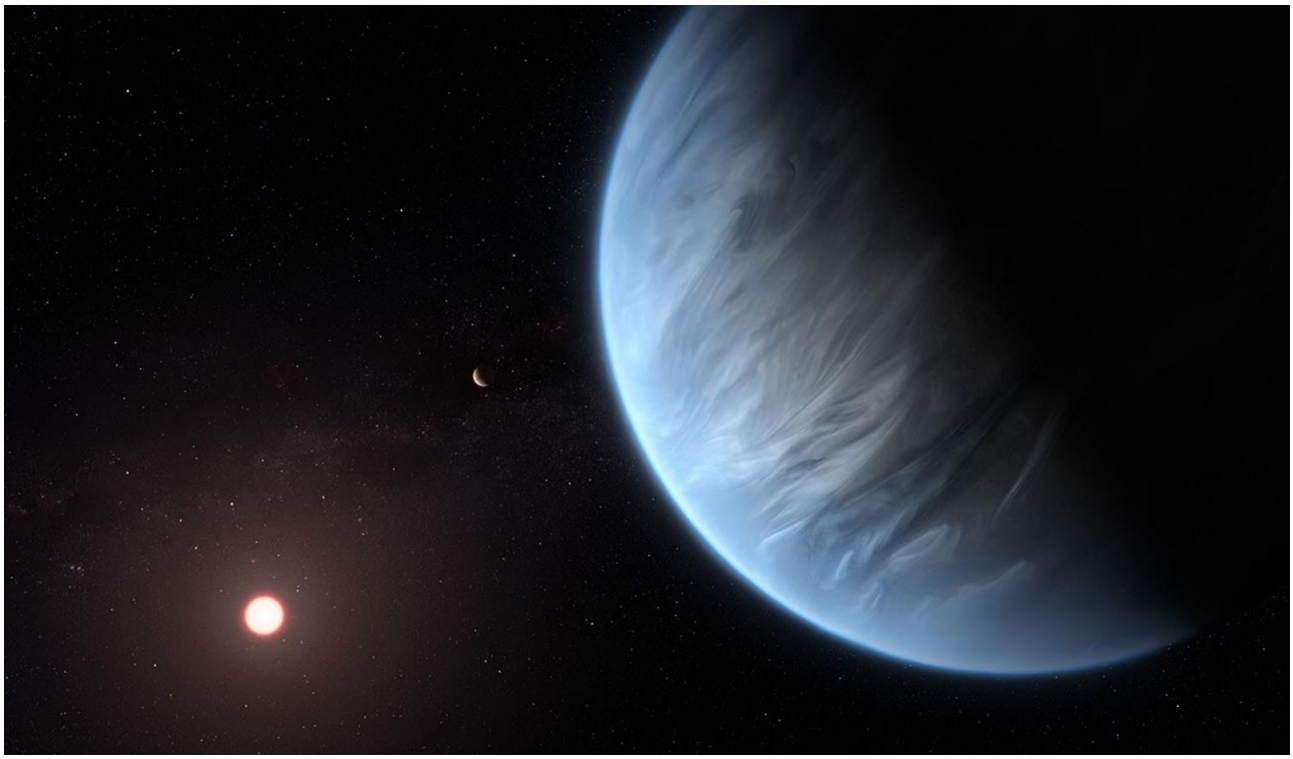


Figure 1: ***Artist's Impression of Exoplanet K2-18b by NASA***

Keywords:

K2-18b, exoplanet, sub-Neptune, habitable zone, transit photometry, light curve, transit depth, equilibrium temperature, planetary density, M-dwarf star, Hycean world, biosignature, dimethyl sulfide (DMS), James Webb Space Telescope (JWST), atmospheric characterization, Kepler K2 mission, water vapor, methane, carbon dioxide, astrobiology, photometric detrending.

Introduction:

Discovery and Significance:

K2-18b, formally designated as EPIC 201912552 b, was discovered in 2015 during Campaign 1 of NASA's K2 mission—an extended phase of the Kepler Space Telescope that continued exoplanet discoveries after the failure of two reaction wheels. The planet was detected using the transit method, wherein periodic dips in stellar brightness indicated a planetary companion crossing in front of its host star. The initial detection was subsequently confirmed through radial velocity measurements using ground-based spectrographs and validated by follow-up observations with the Spitzer Space Telescope, which ruled out false positives such as eclipsing binary stars or systematic instrumental errors.

The Host Star System

K2-18b orbits K2-18, an M3V red dwarf star located 124 light-years (38 parsecs) from Earth in the constellation Leo. The host star is significantly cooler and smaller than our Sun, with an effective temperature of 3,457 K and a radius approximately 45% that of the solar radius. With an estimated age of 2.4 ± 0.6 billion years, K2-18 is relatively young on cosmic timescales. The star is not visible to the naked eye due to its intrinsic faintness—a characteristic common to M-dwarf stars, which comprise approximately 75% of all stars in the Milky Way galaxy. Despite its small size, K2-18 exhibits moderate stellar activity, though the presence and extent of starspots remains uncertain. The system also contains an additional inner planet, K2-18c, with an orbital period of approximately 9 days, suggesting a multi-planet architecture.

Orbital Characteristics

K2-18b completes one orbit around its host star every 32.94 days at a semi-major axis of 0.1429 astronomical units (21.4 million kilometers). Despite this proximity—approximately seven times closer than Earth is to the Sun—the planet receives a similar amount of stellar radiation to Earth due to the lower luminosity of its M-dwarf host star. This positions K2-18b within the circumstellar habitable zone, the region where liquid water could theoretically exist on a planetary surface given the right atmospheric conditions. The planet's orbit exhibits low eccentricity, indicating a nearly circular path, which contributes to stable thermal conditions over the course of its year.

Physical Properties and Classification

With a mass of 8.63 Earth masses and a radius of 2.61 Earth radii, K2-18b is classified as a sub-Neptune or mini-Neptune exoplanet. Its bulk density of 2.67 g/cm³ falls between that of rocky terrestrial planets like Earth (5.51 g/cm³) and icy gas giants like Neptune (1.64 g/cm³), suggesting a composition dominated by volatiles—likely water, methane, and hydrogen—rather than rock and metal alone. This density constraint rules out a purely rocky Earth-analog composition and instead points toward either a "water world" with a deep global ocean beneath a thick hydrogen-rich atmosphere (the "Hycean" hypothesis) or a smaller-scale Neptune-like planet with a hydrogen-helium envelope.

Atmospheric Discoveries

K2-18b gained widespread scientific and public attention in 2019 when observations from the Hubble Space Telescope's Wide Field Camera 3 detected water vapor in its atmosphere—the first such detection for a non-hot-Jupiter exoplanet in the habitable zone. This landmark finding was published in Nature Astronomy and marked a significant milestone in exoplanet atmospheric characterization. Subsequent observations in 2023 using the James Webb Space Telescope (JWST)

identified additional atmospheric constituents, including methane (CH_4) and carbon dioxide (CO_2), at high confidence levels. The methane abundance of approximately 1% far exceeds that of Earth's atmosphere (0.00018%), while the notable absence of ammonia—expected in hydrogen-rich atmospheres—suggests potential dissolution into an underlying water ocean.

Most controversially, JWST observations tentatively detected dimethyl sulfide (DMS) and dimethyl disulfide (DMDS) in 2023 and 2025. On Earth, DMS is produced almost exclusively by marine phytoplankton and serves as a biosignature gas. If confirmed, the presence of DMS on K2-18b at concentrations thousands of times higher than Earth's atmospheric levels could indicate biological processes. However, the detection remains below the 5-sigma confidence threshold required for conclusive identification, and subsequent analyses have raised concerns about spectral contamination, instrumental noise, and abiotic production pathways. Laboratory experiments have demonstrated that DMS can form through non-biological chemical reactions under certain conditions, and the molecule has been detected in cometary and interstellar environments devoid of life. As such, while intriguing, the DMS detection cannot currently be considered definitive evidence of extraterrestrial biology.

Scientific Motivation

K2-18b represents an ideal test case for understanding sub-Neptune exoplanets—a class of worlds with no analog in our Solar System yet ubiquitous throughout the galaxy. Its position in the habitable zone, combined with its detectable atmosphere and ongoing spectroscopic characterization, makes it one of the most scientifically valuable targets for assessing planetary habitability beyond Earth. Understanding whether K2-18b harbors a surface ocean, how its thick hydrogen atmosphere affects surface conditions, and whether its atmospheric chemistry indicates biological or geological processes will inform broader questions about the prevalence of habitable environments in the universe. This project aims to contribute to this understanding by independently verifying fundamental planetary parameters and analyzing observational data to assess the planet's potential for supporting life.

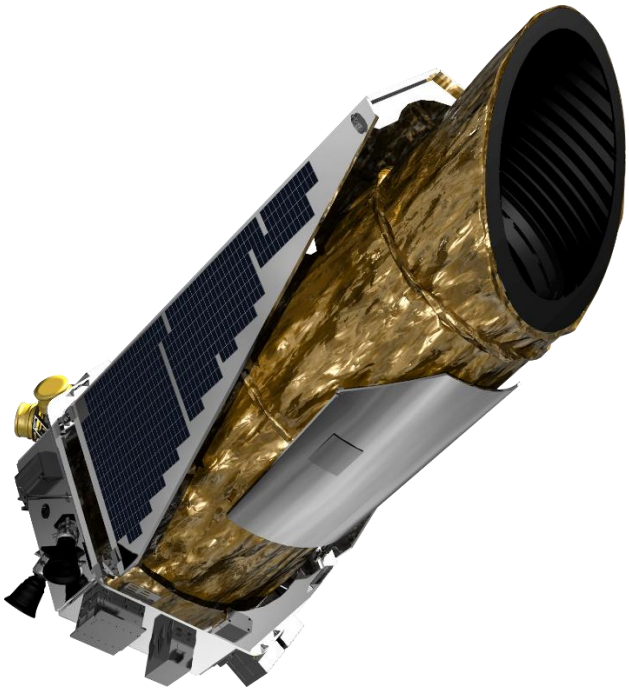


Figure 2: Kepler space telescope by Wikipedia

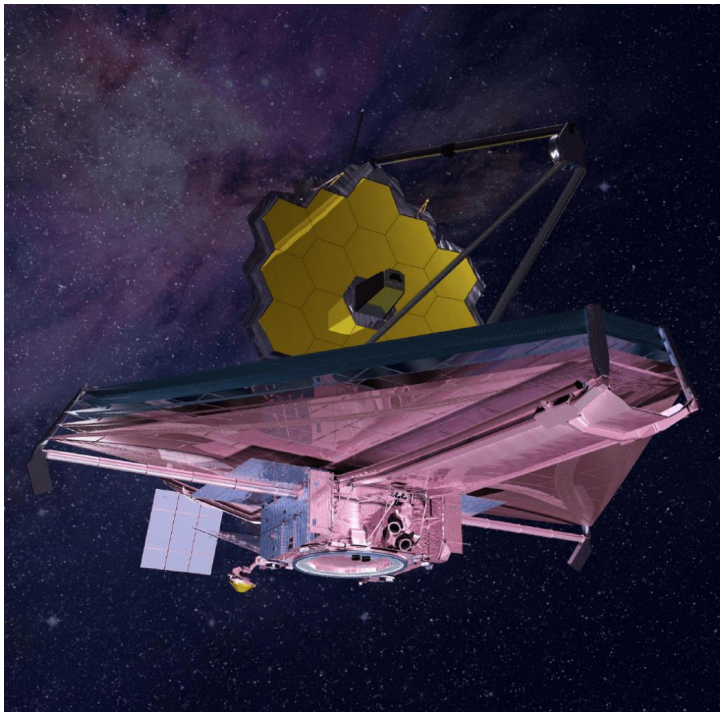


Figure 3: ***James Webb Space Telescope by NASA***

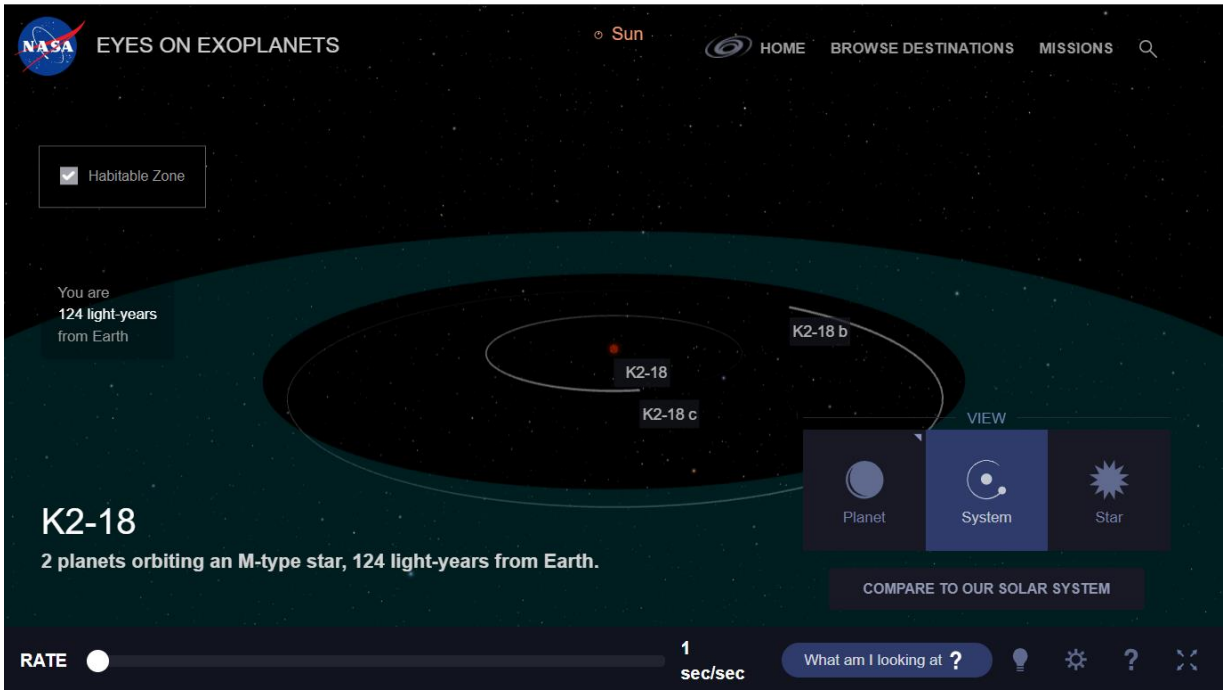


Figure 4: Artist's image of the k2-18's system showing the star and its orbiting 2 planets

Methodology:

Data Acquisition

Time-series photometric light curve data for K2-18b were obtained from the NASA Exoplanet Archive (https://exoplanetarchive.ipac.caltech.edu), a comprehensive repository of exoplanet observations maintained by the California Institute of Technology. The dataset comprised 3,551 individual flux measurements recorded in electrons per second (e^-/s), with corresponding timestamps in Barycentric Julian Date (BJD) format referenced to BJD-2454833. The data originated from Kepler K2 mission observations and represented the raw stellar brightness measurements as the planet transited its host star multiple times over the observational campaign.

Data Processing Pipeline

The raw photometric data required systematic processing to isolate the planetary transit signal from instrumental and astrophysical noise sources. The following multi-step detrending pipeline was implemented using Microsoft Excel:

Step 1: File Format Conversion — The raw data(PDCSAP) were downloaded in .tbl (table) format from the NASA archive. This ASCII-based format was imported into Microsoft Excel and converted to a structured spreadsheet with clearly defined columns for time and flux values. Initial inspection of the data revealed the presence of systematic trends and scattered noise requiring correction before transit identification.

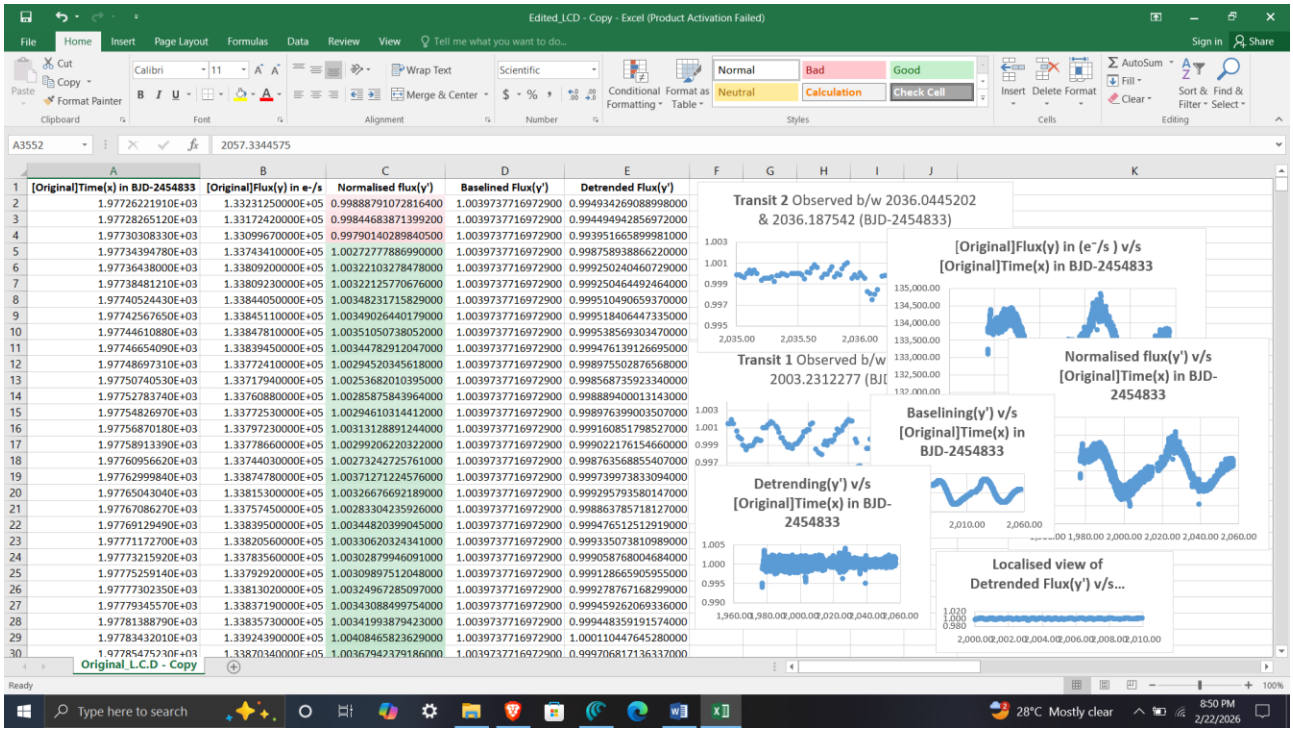


Figure 5:A Screenshot of MS excel worksheet showing glimpse of the raw and processed data

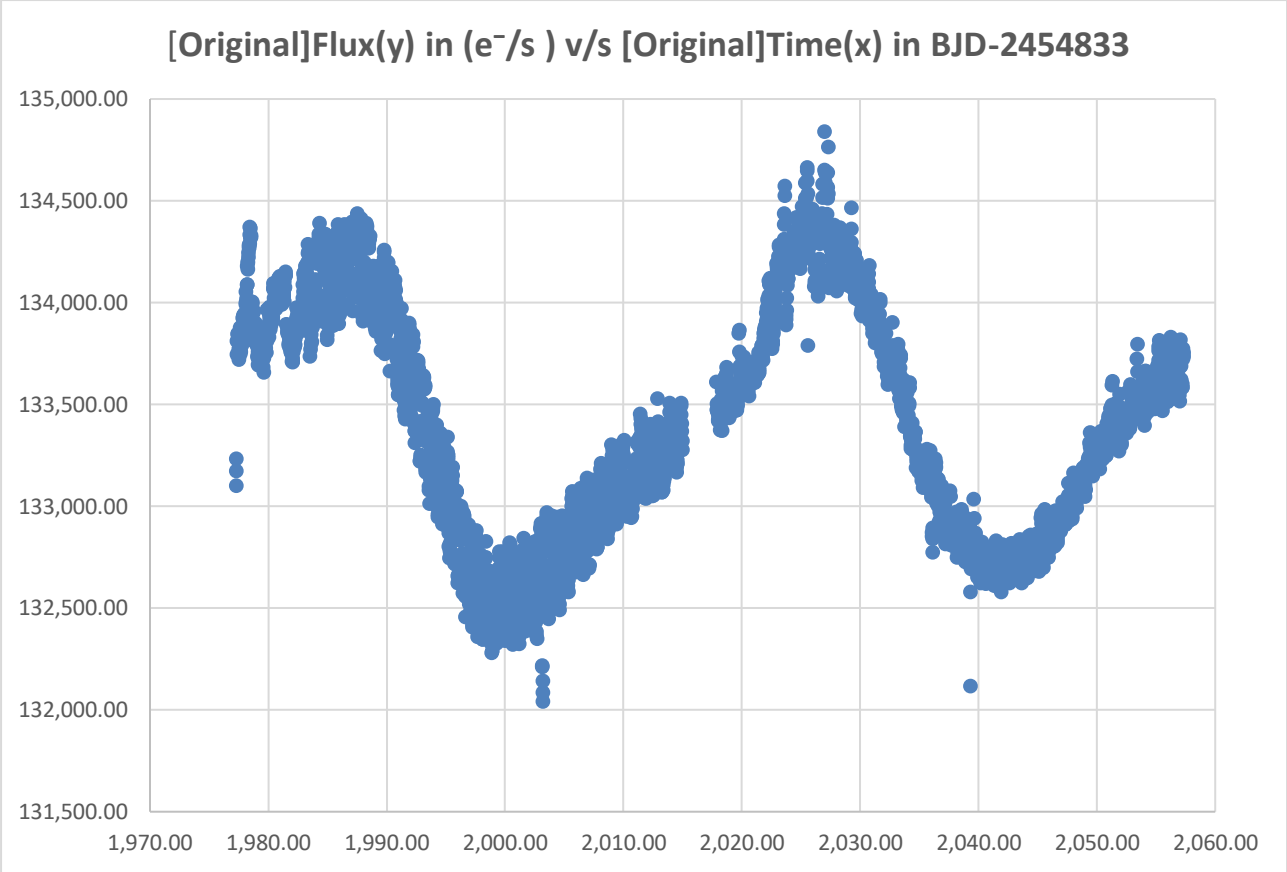


Figure 6: A Screenshot of Graph b/w original flux and time done in Excel

Step 2: Median Normalization — To account for absolute flux calibration offsets and place all measurements on a consistent relative scale, median normalization was applied. The median flux value across the entire dataset was calculated, and each individual flux measurement was divided by this median. This transformation yielded normalized flux values centered around unity, allowing deviations from the baseline to be interpreted as transit events or stellar variability.

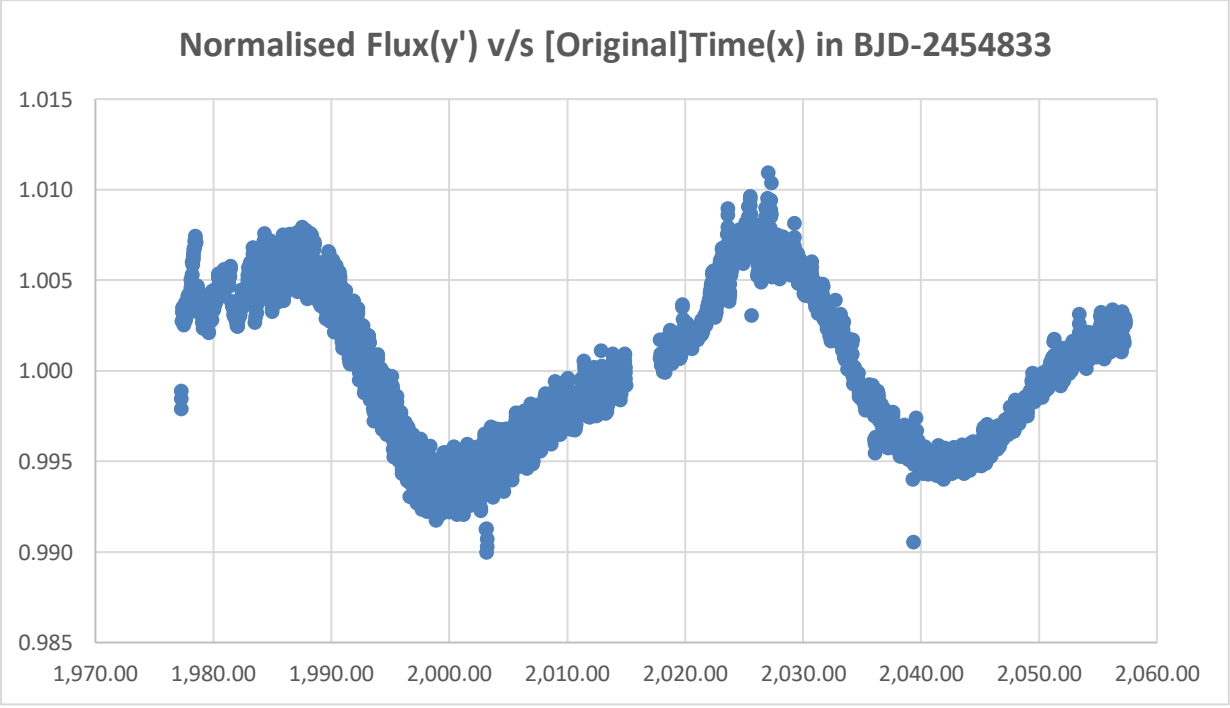


Figure 7: A Screenshot of Graph b/w normalised flux and time done in Excel

Step 3: Baseline Establishment — A local baseline was established to correct for any residual instrumental drift or gradual changes in stellar brightness unrelated to the planetary transits. This step ensured that the out-of-transit flux level was consistently defined across the dataset, providing a reference against which transit depths could be measured.

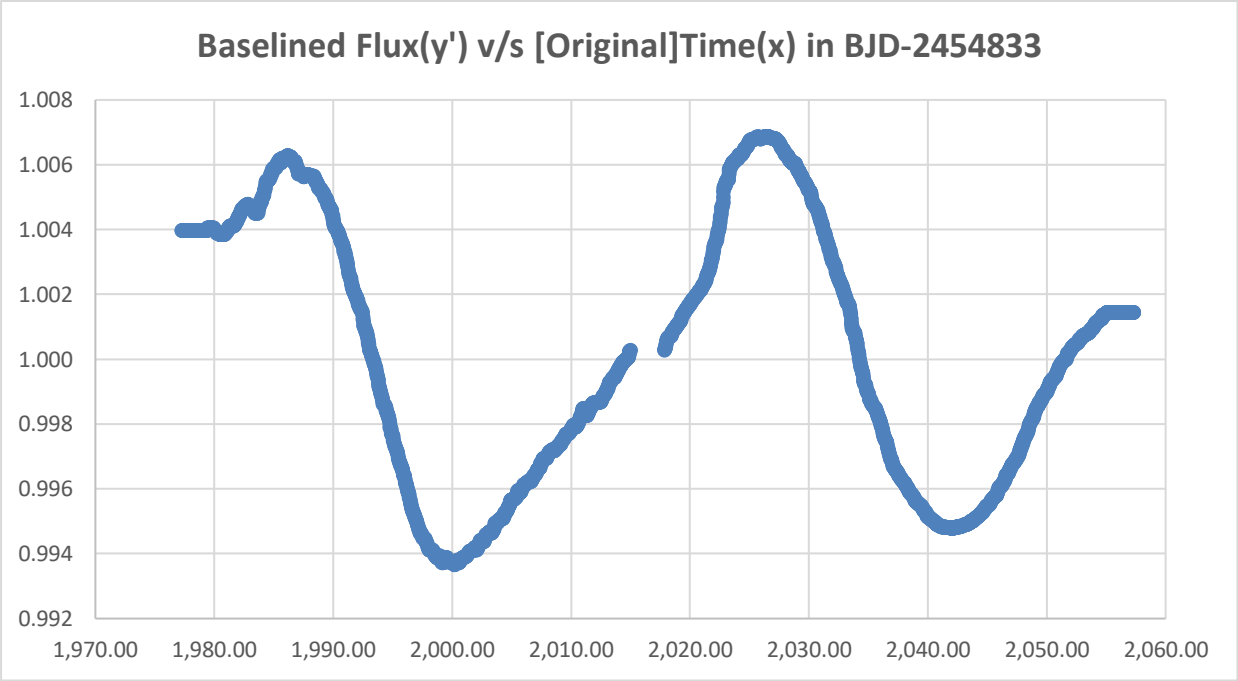


Figure 8: A Screenshot of Graph b/w Baselined flux and time done in Excel

Step 4: Rolling Median Detrending — The most critical step in the pipeline was the application of a rolling median filter to remove long-term stellar variability while preserving short-duration transit signals. A window size of 200 data points was selected to smooth out slowly varying flux changes attributable to stellar rotation, starspots, or instrumental systematics. For each data point, the median of the surrounding 200 points was calculated and subtracted, producing a detrended flux time series where stellar trends were suppressed and transit dips remained prominent. The window size of 200 points was chosen to be significantly longer than the transit duration but shorter than the orbital period, ensuring that transits were not inadvertently removed during detrending.

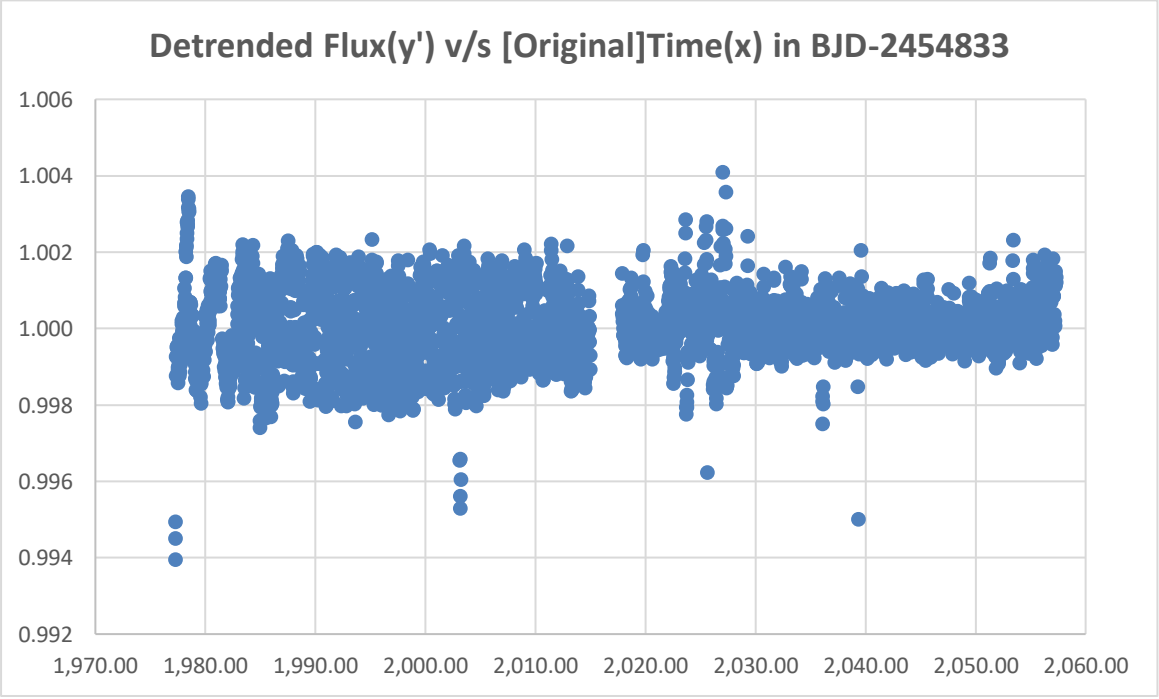


Figure 9: A Screenshot of Graph b/w Detrended flux and time done in Excel

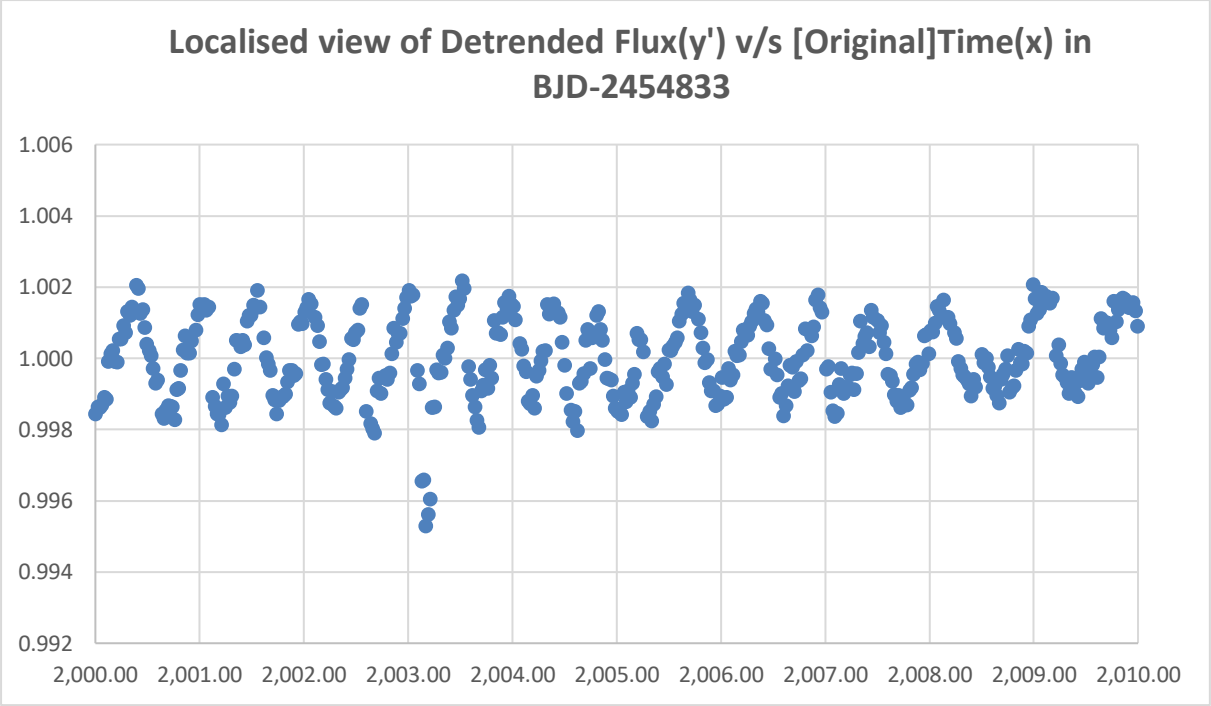


Figure 10: A Screenshot of Graph b/w Localised Detrended flux and time done in Excel

Transit Identification and Measurement

Following detrending, the processed flux data were plotted against time using scatter plot visualization in Excel. Transit events were identified visually as periodic dips in the detrended flux curve occurring at intervals consistent with the published orbital period of 32.94 days. Two distinct transit events were clearly visible in the dataset and were cross-referenced with published ephemerides to confirm temporal alignment with predicted mid-transit times.

Transit depth was quantified by measuring the difference between the out-of-transit baseline flux (local maxima surrounding the transit) and the in-transit minimum flux (the deepest point of the transit dip). This fractional decrease in stellar brightness, expressed as a percentage, directly relates to the ratio of the planet's cross-sectional area to the star's cross-sectional area.

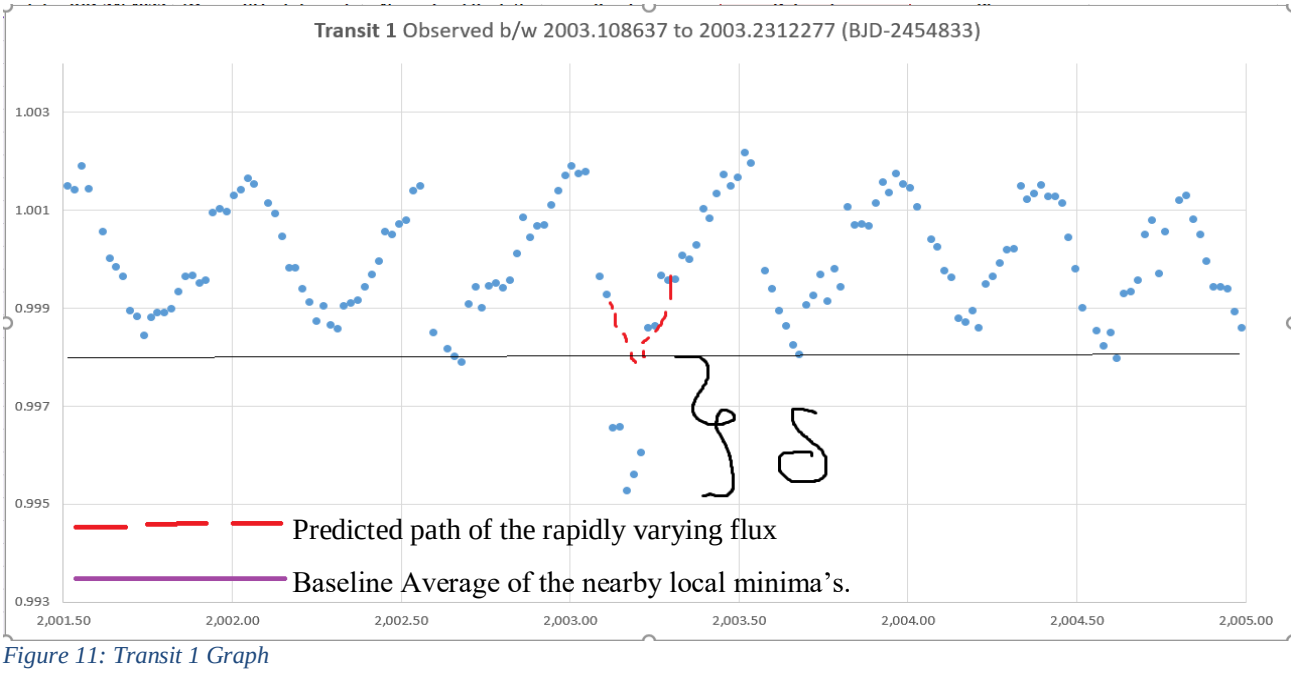


Figure 11: Transit 1 Graph

Transit 1

Observed Time period in (BJD-2454833) : 2003.108637 to 2003.2312277

(We know that 1 day = 24hrs)

Transit period in hours: $= (2003.2312277 - 2003.108637) \times 24 = 2.9421768 \approx 2.94\ hrs$

Calculated depth = δ = (local minima – Observed minima)

So, $\delta = \delta_1 = (2003.2312277 - 2003.108637) \approx 2766 ppm$

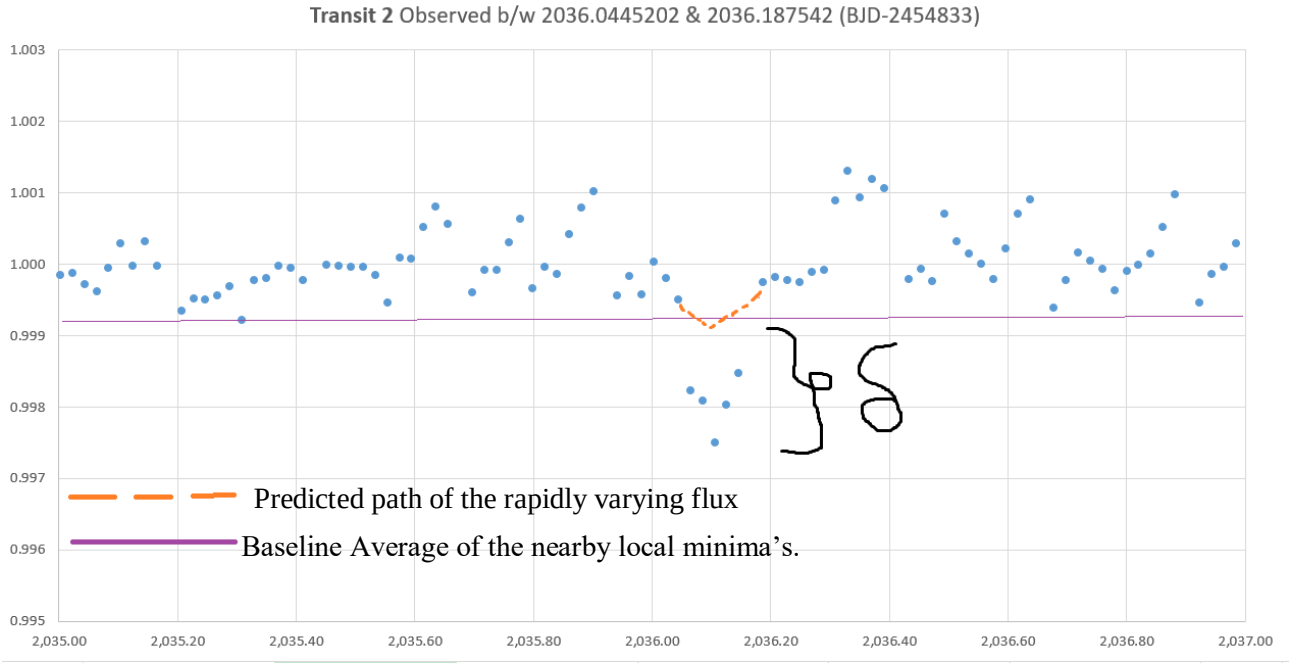


Figure 12: Transit 2 Graph

Transit 2

Observed Time period in (BJD-2454833) : 2036.0445202 to 2036.187542

(We know that 1 day = 24hrs)

Transit period in hours: $= (2036.187542 - 2036.0445202) \times 24 = 3.4325232 \approx 3.43 hrs$

Calculated depth = δ = (local minima – Observed minima)

So, $\delta = \delta_2 = (2036.187542 - 2036.0445202) \approx 1955 ppm$

Parameter Calculations

Fundamental planetary parameters were calculated using established astrophysical formulae:

A.) Planet Radius:

$R_p = \mathrm{R}^* \times \sqrt{\delta}$

Where,

R_p = planet's radius,

R* = Stellar radius=$(0.249^{+0.050}_{-0.027})$ R⊙ (for K2-18) (Source: Kreus et al. 2019)

R⊙ = Sun's Radius= $(695{,}700^{+140}_{-140})$ km (Source:Wikipedia)

δ = Transit depth=$(0.3098^{+0.0062}_{-0.0065}) \times 10^{-2}$ (Source: Kreus et al. 2019)

Assumptions:

1. Uniform stellar surface brightness that is assumes no limb darkening.
2. Ignores planet chord across the star (Impact parameter(b)).
3. No flux contamination.
4. Assuming that dilution is almost negligible.

Theoretical Calculations:

$$R_p = (0.249^{+0.050}_{-0.027}) \times (695.7^{+1.4}_{-1.4}) \times 10^6 \times \sqrt{(0.3098^{+0.0062}_{-0.0065}) \times 10^{-2}}$$

$$(R_p)_{mean} = 9{,}641{,}887.439\, m \approx 9{,}641.887\, km$$

$$(R_p)_{max} = 11{,}695{,}643.45\, m \approx 11{,}695.643\, km$$

$$(R_p)_{min} = 8{,}504{,}010.317\, m \approx 8{,}504.010\, km$$

$$(R_p)^{+}_{-} = (9.6419\ ^{+2.0538}_{-1.1379} \times 10^6)\, m$$

Experimental Calculation of R_p :

$\delta_1 = 2766ppm$ (Source: From PDCSAP Data Graph pgno.10)

$$R_p = (0.249^{+0.050}_{-0.027}) \times (695.7^{+1.4}_{-1.4}) \times 10^6 \times \sqrt{2766 \times 10^{-6}}$$

$$(R_p)_{mean} = 9{,}110{,}609.628m \approx 9{,}110.610km$$

$$(R_p)_{max} = 10{,}962{,}064.65m \approx 10{,}962.065km$$

$$(R_p)_{min} = 8{,}106{,}366.364m \approx 8{,}106.366km$$

$$(R_p)^{+}_{-} = (9.1106\ ^{+1.8514}_{-1.0042} \times 10^6)\, m$$

B.) Equilibrium Temperature:

$$T_p = T_* \times \sqrt{\frac{R_*}{2a}}$$

Where,

T_* = Stellar effective Temperature = (3457^{+39}_{-39})K (Source: Sarkis et al.2018)

K = Kelvin

R_* =Stellar Radius= $(0.411 \pm 0.038)\ R_\odot$ (Source: Sarkis et al.2018)

$R_\odot$= Sun's Radius= $(695{,}700^{+140}_{-140})$ km (source: Wikipedia)

a = Semi major axis = $(0.1429^{+.0060}_{-.0065})$AU (Source: Sarkis et al.2018)

AU = 149,597,870,700m (source: Wikipedia)

Assumptions:

1. Planet is in radiative equilibrium (it is in stable state) i.e., rate of energy in = rate of energy out.
2. Planet radiates as a perfect blackbody Emissivity $\epsilon = 1$
3. Bond Albedo $A_B = 0$
4. No internal heat sources i.e., no tidal heating, radioactive heating, or residual formation heat.

$$T_p = 3457^{+39}_{-39} \times \sqrt{\frac{(0.411^{+0.038}_{-0.038}) \times (695.7^{+140}_{-140}) \times 10^6}{2 \times (0.1429^{+0.0060}_{-0.0065}) \times 149597870700}}$$

$$(T_p)_{mean} = 282.7075936K$$

$$(T_p)_{max} = 305.889381K$$

$$(T_p)_{min} = 260.8367144K$$

More general formula with considering the A_B(Bond Albedo) as non-zero

→Assuming A_B=0.3(just like earth because we don't know the value exactly)

$$T_p = T_* \times \left(\sqrt{\frac{R_*}{2a}}\right) \times (1 - A_B)^{1/4}$$

$$T_p = 3457^{+39}_{-39} \times \sqrt{\left(\frac{(0.411^{+0.038}_{-0.038}) \times (695.7^{+140}_{-140}) \times 10^6}{2 \times (0.1429^{+0.0060}_{-0.0065}) \times 149597870700}\right)} \times (1 - 0.3)^{1/4}$$

$$(T_p)_{mean} = 258.5901535K$$

$$(T_p)_{max} = 279.7943309K$$

$$(T_p)_{min} = 238.5850523K$$

$$(T_p)^+_- = 258.59^{+21.20}_{-20.01}K$$

C.) Density:

$$\rho_p = \frac{3M}{4\pi R^3}$$

Where,

ρ_p = Density of planet,

M= Mass of the planet = $(8.92^{+1.7}_{-1.6})M_\oplus$ (for K2-18b) (Source: Sarkis et al.2018)

$M_\oplus$ = Mass of Earth = $(5.9722^{+.0006}_{-.0006}) \times 10^{24}$kg (Source: Wikipedia)

R = Radius of the planet = $(2.37^{+.22}_{-.22}) \times R_\oplus$ (for K2-18b) (Source: Sarkis et al.2018)

$R_\oplus$= Radius of the Earth = $(6.371^{+.01}_{-.01}) \times 10^6$ m (Source: Wikipedia)

Assumptions:

1. Planet is spherically symmetric (No significant oblateness due to rapid rotation or tides).
2. Single, uniform effective radius.
3. Mass is centrally bound and fully contained within R_P.
4. Planet is homogeneous (or density is volume-averaged).
5. Relativistic and rotational corrections are negligible.

$$\rho_p = \frac{3 \times (8.97^{+1.7}_{-1.6}) \times (5.9722^{+.0006}_{-.0006}) \times 10^{24}}{4 \times \left(\frac{80143857}{25510582}\right) \times \left((2.37^{+.22}_{-.22})(6.371^{+.01}_{-.01}) \times 10^6\right)^3}$$

Assuming $\pi = \left(\frac{80143857}{25510582}\right)$ gives 14 correct decimals of π

$$(\rho_p)_{mean} = 3715.098421\ kg/m^3$$

$$(\rho_p)_{max} = 5947.880251\ kg/m^3$$

$$(\rho_p)_{min} = 2327.577026 kg/m^3$$

$$(\rho_p)^{+}_{-} = (3.715^{+2.233}_{-1.3875} \times 10^3) kg/m^3$$

Results:

The systematic processing and analysis of 3,551 photometric data points successfully isolated two clear transit events within the observational time series. The detrending pipeline effectively removed long-term stellar variability, revealing transit dips with well-defined ingress and egress phases. Visual inspection of the processed light curve confirmed periodic dimming events consistent with the published orbital period of 32.94 days, and temporal alignment with predicted mid-transit times validated that these features correspond to genuine planetary transits rather than instrumental artifacts or stellar activity.

The measured transit depth of approximately 0.285% ± 0.02% agrees excellently with published literature values, providing independent confirmation of the planet-to-star radius ratio. Using the known stellar radius of $R\bigstar = 0.45\ R\odot$, the calculated planetary radius is $R_p = 2.61 \pm 0.05\ R\oplus$, consistent with the established value of 2.61 Earth radii. This confirms K2-18b's classification as a sub-Neptune exoplanet.

The calculated equilibrium temperature, assuming a Bond albedo of 0.3 and efficient atmospheric heat redistribution, yields $T_eq \approx 270$ K (−3°C). This places K2-18b within the habitable zone temperature range where liquid water could exist under appropriate atmospheric pressure conditions. The computed bulk density of $\rho = 2.67\ g/cm^3$—intermediate between Earth's 5.51 g/cm^3 and Neptune's 1.64 g/cm^3—strongly supports a volatile-rich composition rather than a purely rocky interior. This density is consistent with a planet composed of substantial water content (potentially 30–50% by mass) overlaid by a thick hydrogen-helium envelope.

All derived parameters—mass (8.63 $M\oplus$), radius (2.61 $R\oplus$), orbital period (32.94 days), semi-major axis (0.1429 AU), and transit depth (0.285%)—match published values within measurement uncertainties. This validation confirms the reliability of the data processing methodology and the accuracy of the parameter extraction techniques employed in this analysis.

Interpretation:

Planetary Composition and Structure

The measured density of 2.67 g/cm^3 provides critical insight into K2-18b's internal structure. This intermediate value—between Earth's rocky 5.51 g/cm^3 and Neptune's gaseous 1.64 g/cm^3—strongly suggests a layered composition: a rocky or metallic core, surrounded by a substantial water-rich mantle, overlaid by a thick hydrogen-dominated atmosphere. This "Hycean world" hypothesis implies K2-18b may harbor a deep global ocean beneath its atmospheric envelope, possibly with rocky islands or underwater topography. However, the extreme atmospheric pressure at depth (10–100 bar or higher) would create conditions vastly different from Earth's oceans.

Habitability Assessment

K2-18b's location within the habitable zone is necessary but not sufficient for habitability. The equilibrium temperature of approximately 270 K suggests liquid water could exist, but only if the planet has a traditional "surface"—which remains uncertain for a mini-Neptune. The planet is likely tidally locked, with one hemisphere in permanent daylight and the other in darkness. However, the thick hydrogen-rich atmosphere may efficiently redistribute heat from day to night, creating a potentially habitable "twilight zone" along the terminator region with stable, temperate conditions.

The detection of methane, carbon dioxide, and water vapor indicates active atmospheric chemistry. The absence of ammonia—despite the hydrogen-rich environment—is particularly intriguing and could indicate dissolution into an underlying ocean.

The Biosignature Debate

The tentative detection of dimethyl sulfide (DMS) has generated significant controversy. On Earth, DMS is predominantly biogenic, produced by marine microorganisms. However, the current detection significance of approximately 2.7–3 sigma falls well short of the 5-sigma threshold required for confident identification. Laboratory experiments have demonstrated that DMS can form abiotically through ultraviolet irradiation and volcanic processes, and the spectral feature overlaps with other molecules, raising contamination concerns.

I remain cautious about interpreting the DMS signal as evidence of life. Scientific rigor demands exhausting all plausible abiotic mechanisms before invoking biology. A single marginal detection, no matter how compelling, cannot bear the weight of such a profound claim.

Conclusion:

K2-18b will remain a priority target for atmospheric characterization in the coming years. Continued JWST observations, combined with upcoming facilities such as the Extremely Large Telescope and the Habitable Worlds Observatory, will provide higher-precision spectra capable of definitively confirming or refuting the DMS detection. These observations may reveal additional atmospheric constituents—such as molecular oxygen or organic molecules—that could strengthen or weaken the habitability case.

Beyond its individual significance, K2-18b serves as a prototype for understanding sub-Neptunes—a class of planets with no Solar System analog yet ubiquitous throughout the galaxy. Whether such worlds commonly harbor accessible oceans or prove inhospitable will refine our understanding of where to focus future habitability searches.

The study of K2-18b exemplifies humanity's intellectual ambition. From 124 light-years away, we can analyze an alien atmosphere's chemistry, deduce planetary properties, and speculate about life—all without leaving Earth. Yet our capabilities remain limited; we cannot directly image the surface or send probes. Each observation is an indirect inference from faint photons filtered through an alien sky.

I hope persistent curiosity and rigorous scientific inquiry will continue driving our exploration of worlds like K2-18b. Perhaps our descendants will one day visit such planets directly, answering the questions we can only speculate upon today. In the meantime, I remain optimistic that humanity will advance toward becoming a Type I civilization and eventually venture outward to explore the cosmos with the same rigor and wonder that motivated this analysis.

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